The scrjrnl class

Raphaël Pinson
raphink@gmail.com

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1 Introduction

The scrjrnl class is based of the scrbook class, and inherits all of its commands and settings. Please refer to the Koma-Script documentation for the detail of all of these. Note that the scrjrnl class is not an official Koma-Script class.

2 Usage

```
\documentclass[<options>]{scrjrnl}
```

2.1 Options

The class options are described below.

`monthpages`

By default, the month pages are skiped and only the days are typeset. If you wish to display a page for each starting month, use this setting.

`fancytabs`

This class can automatically make use of the fancytabs package to typeset tabs on the side of every odd page, with the name of the current month.

2.2 The `journal` environment

The `journal` environment is the main addition of this class. It provides an environment in which to typeset journal entries.

The `journal` environment takes the following options:

`startmonth`

The month in which to start the journal. Defaults to 1.

`startday`

The day in which to start the journal. Defaults to 1.

2.3 Typesetting months

Once the `journal` environment in initiated, you can use the `jrnlmonth` and `jrnlday` commands to typeset journal entries.

`\jrnlmonth`

The `\jrnlmonth` command starts a new month. It takes the following options:

`month`

Forces the current month instead of simply incrementing from the previous one. This parameter is an integer between 1 and 12.

`startday`

Start month at a given day, instead of 1.

2.4 Typesetting a journal entry

`\jrnlday`

In order to typeset a single journal entry, use the `\jrnlday` command. This command takes a mandatory argument which is the journal entry's title, as well as the following options:

`day`

Forces the current day instead of simply incrementing from the preivous one. This parameter is an integer.

`ante`

Add a text before the date.

`post`

Add a text after the date.

2.5 Index

When typesetting a journal, you might want to rely only on day entries to relate to pages and turn off completely page numbers by using an `empty` page style.

In this situation, if you have to create an index, you might want index entries to link to dates in the journal rather than page numbers.

The scrjrnl class records each typeset day in macros so they can be used to generate an index. The macros are named `jrnldaydate\thejrnlday`, with the page day corresponding to the entry, numbered from the first typeset day in the journal.

Below is an example of an index generated using the bibleref package, and linking to journal entries instead of pages:

```
% Index by day
\renewcommand*{\bvidxpgformat}{transform}

% Get the date for the recorded day
\newcommand\getdaydate[1]{\csname jrnldaydate#1\endcsname}

% Go through index items
\newcommand{\transform}[1]{%
  \def\secondparam{0}%
  \forcsvlist\decodesec{#1}}
\newcommand{\decodesec}[1]{%
  \ifthenelse{\secondparam=1}{, }{}%
  \mbox{\textsc{\getdaydate{#1}}}
  \def\secondparam{1}}

% Patch imakeidx to map journal entry instead of page
\makeatletter
\patchcmd{\@index}{\@wrindex}{\@Wrindex}{}{}
\let\@Wrindex\@wrindex
\patchcmd{\@Wrindex}{\thepage}{\thejrnltotalday}{}{}
```

```
\let\protected@iwrite\protected@write
\patchcmd{\protected@iwrite}{\write}{\immediate\write}{}{}
\patchcmd{\@Wrindex}{\protected@write}{\protected@iwrite}{}{}
\makeatother
```

3 Implementation

```
\ProvidesClass{scrjrnl}
\DeclareOption{scrartcl}{\def\jrnl@class{scrartcl}}
\DeclareOption{scrreprt}{\def\jrnl@class{scrreprt}}
\DeclareOption{scrbook}{\def\jrnl@class{scrbook}}
\DeclareOption*{\PassOptionsToClass{\CurrentOption}{\jrnl@class}}
\ExecuteOptions{scrbook}
\newif\if@jrnl@monthpages
\DeclareOption{monthpages}{\@jrnl@monthpagestrue}
\newif\if@jrnl@fancytabs
\DeclareOption{fancytabs}{\@jrnl@fancytabstrue}
\newif\if@jrnl@cleardaypages
\DeclareOption{cleardaypages}{\@jrnl@cleardaypagestrue}
\ProcessOptions\relax
\LoadClass{\jrnl@class}
\RequirePackage{titlesec}
```

Load datetime after babel

```
\AfterPackage{babel}{\RequirePackage{datetime}}
\newcommand\jrnlformatdate\formatdateny
\newcommand\jrnlformatindexdate\formatdateny
```

More datetime

```
% formatdate without year
\def\formatdateny{\csname noyear\languagename\expandafter\endcsname\formatdate}

\def\noyearenglish#1, \the\@year{#1}
\let\noyearamerican\noyearenglish
\let\noyearbritish\noyearenglish
\def\noyearfrench#1\space\number\@year{#1}
\let\noyeargerman\noyearfrench
\let\noyearaustrian\noyeargerman
\let\noyearswedish\noyearfrench
\let\noyearbreton\noyearfrench
\def\noyearrussian#1\ \number\@year~\cyrg.{#1}
\def\noyearspanish#1\ de~\number\@year{#1}
\let\noyearcatalan\noyearspanish
\def\noyearbasque#1\number\@year.eko\space{#1}
```

Set default lengths

```
\newlength{\jrnldaysepskip}
\setlength{\jrnldaysepskip}{0pt}
\newlength{\jrnldayafterskip}
\setlength{\jrnldayafterskip}{2\baselineskip}
```

Load fancytabs

```
%%\if@jrnl@fancytabs
\RequirePackage{fancytabs}[2012/09/24]
%%\fi
```

Month names

```
\newcommand{\jrnl@monthname}[1]{\monthname[#1]}
\newcommand{\jrnl@curmonth}{\jrnl@monthname{\thejrnlmonth}}
```

Setup counters

```
\newcounter{jrnlstartchapter}
```

```
\newcounter{jrnlday}
\newcounter{jrnlmonth}
\newcounter{jrnlyear}
\newcounter{jrnltotalday}
```

HTML hooks for TeX4HT

```
\newcommand{\HTMLchapHook}[1]{}
\newcommand{\HTMLsecHook}[2]{}
```

Define jrnlmonth

```
\newcounter{jrnlmonthstartday}
\define@key{jrnlmonth}{month}{\setcounter{jrnlmonth}{#1}\addtocounter{jrnlmonth}{-
1}}
\define@key{jrnlmonth}{startday}{\setcounter{jrnlmonthstartday}{#1}\addtocounter{jrnlmonthstar
1}}

\newcommand{\jrnlmonth}[1][]{%
  % Use \thejrnlstatday if first month of journal
  \setcounter{jrnlmonthstartday}{\thejrnlstartday}%
  \setcounter{jrnlstartday}{0}%
  \setkeys{jrnlmonth}{#1}%
  \stepcounter{jrnlmonth}%
  \HTMLchapHook{\jrnl@curmonth}%
  \if@jrnl@monthpages
    \chapter{\jrnl@curmonth}%
    \cleardoublepage%
  \else
    \refstepcounter{chapter}%
    \@maybeautodot\thechapter
    \addchaptertocentry{\thechapter}{\jrnl@curmonth}%
  \fi
  \setcounter{jrnlday}{\thejrnlmonthstartday}%
}
```

Define jrnlday

```
\newcommand{\jrnl@pagedate}{%
  \jrnlformatdate{\thejrnlday}{\thejrnlmonth}{\thejrnlyear}%
}
\define@key{jrnlday}{day}{\setcounter{jrnlday}{#1}\addtocounter{jrnlday}{-
1}}
\define@key{jrnlday}{ante}{\def\jrnlday@ante{#1}}
\define@key{jrnlday}{post}{\def\jrnlday@post{#1}}

\newcommand{\jrnlday}[2][]{%
  \def\jrnlday@ante{}%
  \def\jrnlday@post{}%
  \setkeys{jrnlday}{#1}%
  \stepcounter{jrnlday}%
  \stepcounter{jrnltotalday}%
  \if@jrnl@cleardaypages
  \if@openright\cleardoublepage\else\clearpage\fi
  \fi
  \newpage\section{#2}%
  % TODO: provide index macros to be used with jrnlpagedate\thepage
  \expandafter\xdef\csname jrnldaydate\thejrnltotalday\endcsname{%
    \jrnlday@ante\noexpand\jrnlformatindexdate{\thejrnlday}{\thejrnlmonth}{\thejrnlyear}\jrnlc
}%
  \HTMLsecHook{\jrnlday@ante\jrnlformatdate{\thejrnlday}{\thejrnlmonth}{\thejrnlyear}\jrnlday@
}
```

Define fonts

```
\newcommand\dayheadfont{}
\newcommand\monthheadfont{}
```

Chapter/section styles

```
\newcommand\jrnl@titlestrut{\vrule height 25pt width0pt\relax}
\newcommand{\jrnl@secstyle}{\filcenter\dayheadfont\huge}
\newcommand{\jrnl@seclblstyle}{\raggedright\normalfont\scshape\Large}
\newcommand{\jrnl@seclbl}{\jrnl@seclblstyle\jrnlday@ante\jrnl@pagedate\jrnlday@post\\}

\newcommand{\jrnl@chapstyle}{\filcenter\monthheadfont\Huge\textsc}
```

Define journal environment

```
\newcounter{jrnlstartmonth}
\newcounter{jrnlstartday}
\define@key{jrnl}{startmonth}{\setcounter{jrnlstartmonth}{#1}\addtocounter{jrnlstartmonth}{-1}}
\define@key{jrnl}{startday}{\setcounter{jrnlstartday}{#1}\addtocounter{jrnlstartday}{-1}}
\newlength{\jrnl@bleed}
\define@key{jrnl}{bleed}{\setlength{\jrnl@bleed}{#1}}

\newenvironment{journal}[1][]%
{%
\setcounter{jrnlstartmonth}{0}%
\setcounter{jrnlstartday}{0}%
\setkeys{jrnl}{#1}%
\setcounter{jrnlstartchapter}{\thechapter}
\addtocounter{jrnlmonth}{\thejrnlstartmonth}

\titleformat{\chapter}[block]
  {\vfill}
  {}{0pt}
  {\jrnl@chapstyle}
  [\vfill\vfill]

\if@jrnl@fancytabs
  % We know \paperheight and bleed
  % Calculation base:
  % 6*fancytabsHeight+2*fancytabsTop+5*fancytabsGap = paperheight
  % fancytabsHeight = 10*fancytabsGap
  % fancytabsTop = 5*fancytabsGap + 2*bleed

  % Calculate \calcfancytabsGap as (\the\paperheight-2*bleed)/75
  \newlength{\calcfancytabsGap}
  \pgfmathsetlength{\calcfancytabsGap}{\the\paperheight-2*\the\jrnl@bleed)/75}
  \fancytabsGap{\the\calcfancytabsGap}

  % Calculate \calcfancytabsHeight as 10*\calcfancytabsGap
  \newlength{\calcfancytabsHeight}
  \pgfmathsetlength{\calcfancytabsHeight}{10*\calcfancytabsGap}
  \fancytabsHeight{\the\calcfancytabsHeight}

  % Calculate \calcfancytabsTop as (5*\calcfancytabsGap+2\jrnl@bleed)
  \newlength{\calcfancytabsTop}
  \pgfmathsetlength{\calcfancytabsTop}{5*\calcfancytabsGap+2*\the\jrnl@bleed}
  \fancytabsTop{\the\calcfancytabsTop}
```

```
144
145   % Calculate \fancytabsWidth, without bleed
146   \newlength{\calcfancytabsWidth}
147   \pgfmathsetlength{\calcfancytabsWidth}{\the\paperwidth/12}
148
149   % Calculate \fancytabsWidthBleed, with bleed
150   \newlength{\calcfancytabsWidthBleed}
151   \pgfmathsetlength{\calcfancytabsWidthBleed}{\the\calcfancytabsWidth+\the\jrnl@bleed}
152   \fancytabsWidth{\the\calcfancytabsWidthBleed}
153
154   \fancytabsTextVPos{0.5}
155   % Put text more inside
156   % \fancytabsTextHPos = (\fancytabsWidth/2)/\fancytabsWidthBleed
157   \pgfmathsetmacro{\calcfancytabsTextHPos}{(\the\calcfancytabsWidth/2)/(\the\calcfancytabsWidt
158   \fancytabsTextHPos{\calcfancytabsTextHPos}
159
160   \newcommand{\jrnl@fancytab}{%
161     \ifthenelse{\isodd{\thepage}}%
162       {\fancytab{\jrnl@curmonth}{\thejrnlmonth}}%
163       {}%
164   }
165 \else
166   \newcommand{\jrnl@fancytab}{}%
167 \fi
168
169 \titleformat{\section}[display]
170   {\jrnl@secstyle} % format
171   {\jrnl@seclbl} % label
172   {\jrnldaysepskip} %sep
173   {\jrnl@fancytab%
174    \jrnl@titlestrut\filcenter} %before
175   [\vspace{\jrnldayafterskip}] %after
176
177 \titlespacing{\section}{0pt}{*}{-3pt}
178
179 } % end of \begin{devotional}
180 {} % end of \end{devotional}
```